

Group 2
ARCA

NJIT Campus Buddy

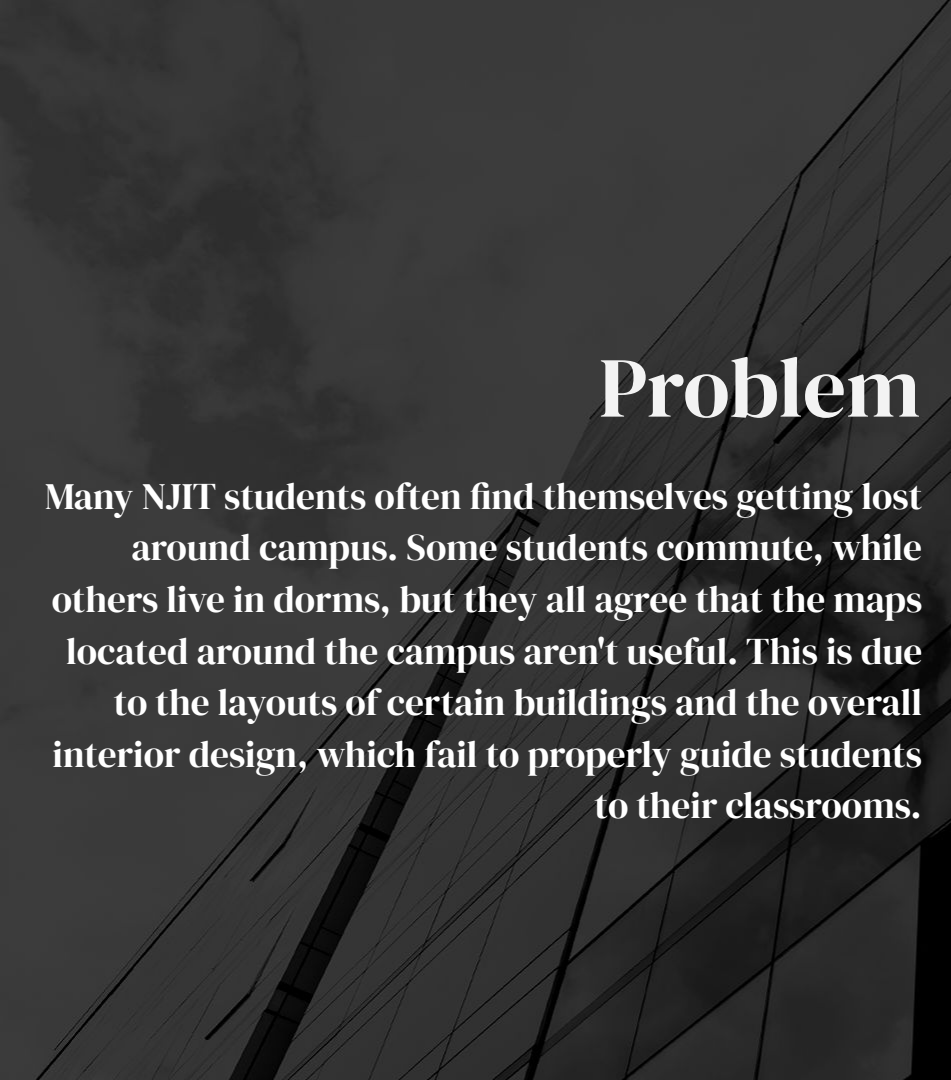
Group Members: Amy Ruiz, Ryan Ramos, Christopher Romero, Anabhayan Ahruran



A low-angle, black and white photograph of several skyscrapers reaching towards a dark sky. The perspective creates a sense of height and scale. A white rectangular frame is centered on the image, containing the text.

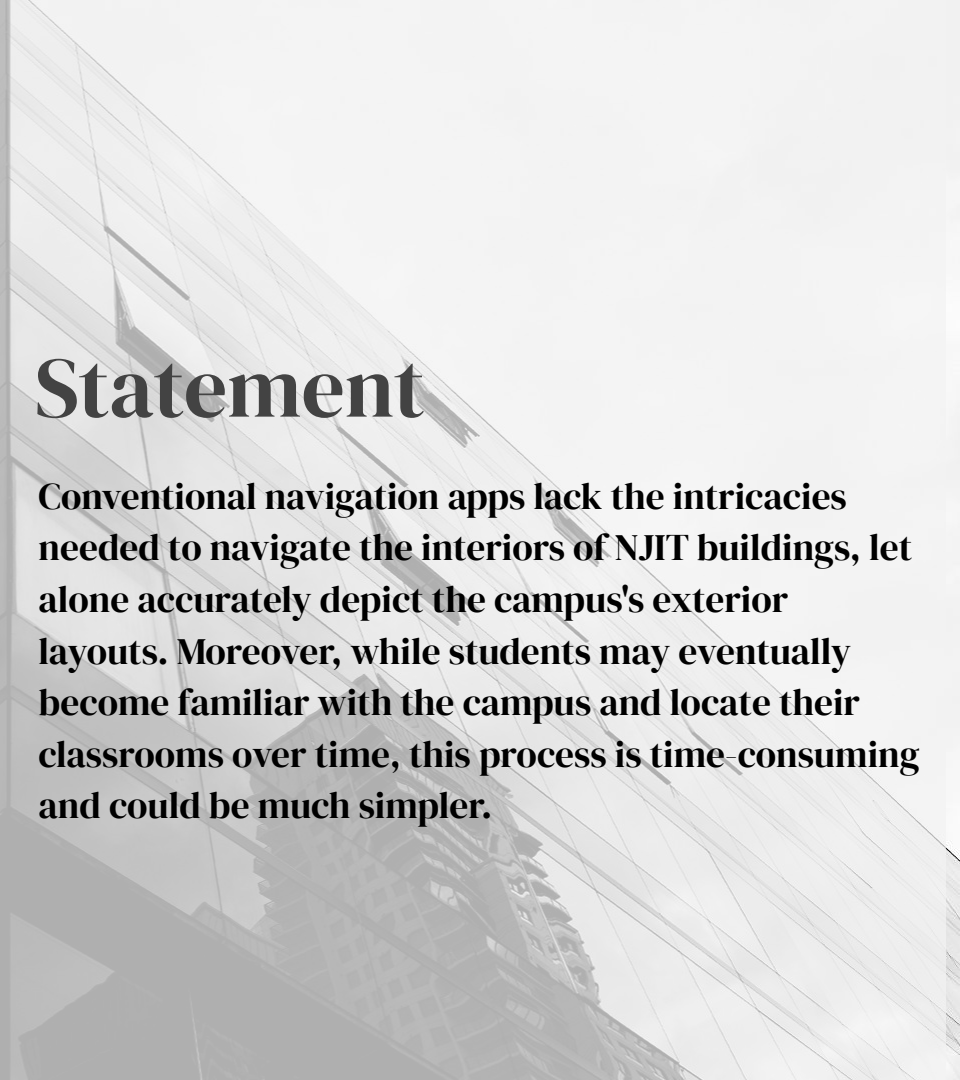
01

Problem
Statement



Problem

Many NJIT students often find themselves getting lost around campus. Some students commute, while others live in dorms, but they all agree that the maps located around the campus aren't useful. This is due to the layouts of certain buildings and the overall interior design, which fail to properly guide students to their classrooms.



Statement

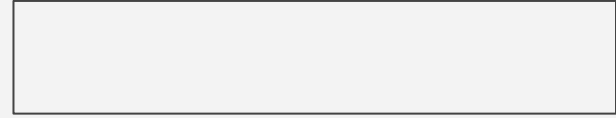
Conventional navigation apps lack the intricacies needed to navigate the interiors of NJIT buildings, let alone accurately depict the campus's exterior layouts. Moreover, while students may eventually become familiar with the campus and locate their classrooms over time, this process is time-consuming and could be much simpler.

A low-angle, black and white photograph of several skyscrapers reaching towards a dark, cloudy sky. The perspective creates a sense of height and scale. The buildings are silhouetted against the lighter sky, with some windows reflecting light.

02

Proposal

High Level Vision / Proposed Solution



Navigation around the NJIT campus poses challenges, especially for first-year students unfamiliar with the layout. Existing solutions like the on-campus map and Google Maps provide limited assistance, often leaving students struggling to locate both buildings and classes.

NJIT Campus Buddy is a platform designed to simplify campus navigation. It offers comprehensive information on class locations, real-time updates on schedule changes, and easy event registration through your UCID. With NJIT Campus Buddy, students can efficiently locate classes, stay informed about schedule changes, and earn points for attendance, redeemable for various campus services. Overall, NJIT Campus Buddy is an all-in-one solution for navigating campus efficiently and staying up-to-date with schedules and events.



A low-angle, black and white photograph of several skyscrapers reaching towards the sky. The perspective creates a sense of height and scale. The buildings are dark, with some windows reflecting light. The sky is a uniform dark grey.

03 Stakeholders

Stakeholders

1. **First-Year/Transfer Students:**
 - *Reason:* They are the primary users of the navigation app, relying on it to find their way around campus and locate buildings and classrooms. The app directly impacts their ability to navigate the campus efficiently, ensuring they arrive at classes on time and don't miss important parts of lectures.
2. **Current NJIT Students:**
 - *Reason:* Even though they may be familiar with the campus layout, the app can still benefit them by providing real-time updates, alternative routes, and information about any changes in classroom locations. This group is indirectly affected as the app enhances overall campus navigation for all students.
3. **NJIT Faculty and Staff (Ex. | Professors):**
 - *Reason:* Faculty members are indirectly impacted as the app can contribute to improved punctuality and attendance, ensuring that students arrive at classes on time. It may also reduce interruptions caused by late arrivals, creating a more conducive learning environment.
4. **NJIT Administration (Ex. | Student senate):**
 - *Reason:* The administration is concerned with the overall campus experience and student success. The navigation app can contribute to a positive student experience by addressing design failures in the campus layout, potentially improving retention rates and overall satisfaction among students.

A low-angle, black and white photograph of several skyscrapers reaching towards the sky. The perspective creates a sense of height and scale. The buildings are dark, with some windows reflecting light. The sky is a uniform dark grey.

04

Research
Insights

Navigation

Has a very hard time finding GITC

Thinks the Architecture building is hard to traverse.

Finds room numbering in Faculty Memorial Hall particularly confusing

Finds the Tiernan and Memorial Hall confusing due to their layouts

Only familiar with outside campus maps

their only help in navigating around NJIT is the on campus map, however, this doesn't help guide them to classes

Wants to be able to see friends location on campus

Experience

Is a second year student

Second year student.

Has basically zero problems getting to class.

is a first year student and it is his first semester here at NJIT

Uses email for notification, doesn't see the need for anything else.

Not typically late for class -- only forgot once.

believes our application would be of value for future semesters.

Desired Features

Would find it helpful for navigating to classes.

Wants to see status updates on the elevators working.

Appreciates accessibility tools (high contrast for blind people, audio for Very Blind people.)

it is very important for him to use an app that is easy and accessible to you

something the user believed we should consider is how well the application would work without internet, as at times the NJIT wifi isn't reliable

would find it very helpful if there is any delays in traffic or classes on way to NJIT

the user would find it nice to get reminders about upcoming events since emails don't serve as great reminders

Wants notifications on campus delays/weather.

App Usage

Knows what the Corq app is. Doesn't use/care about it.

Wants accessibility features like text-to-speech for blind people.

Doesn't use any current navigation apps around campus

Doesn't use apps for anything... Navigation or events.

would love getting a notification when there is an event or class

Would find the app more useful than checking canvas

Is very comfortable with using apps for navigation

rarely uses phone for on-campus activities

Research Insights

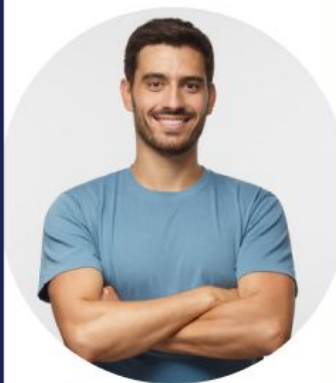
- **Navigation**
 - Certain buildings cause more difficulty in navigation than others.
 - Students typically only use the available campus maps posted outside for navigation.
- **Experience**
 - Majority of students interviewed were first-years and sophomores
 - Most are used to to the current state of navigation.
 - Not typically late for class, but would appreciate an app for finding things sometimes.
- **Desired Features**
 - Wants to see accessibility features for blind/colorblind people.
 - Wants to see realtime status updates for things on campus.
 - Would appreciate notification reminders, and easy to use quickly.
- **App Usage**
 - Knows the apps currently offered by NJIT, but don't use them very often.
 - Comfortable using phones, but there's just no use for navigation purposes yet.

A low-angle, black and white photograph of several skyscrapers reaching towards a dark, cloudy sky. The perspective creates a sense of height and scale. A white rectangular frame is superimposed over the center of the image.

05 Persona

PERSONA

Cameron Anderson



AGE	19
COLLEGE	NJIT
MAJOR	Computer Science
OCCUPATION	Full-time college student
LOCATION	Kearny, NJ
GOAL	To become comfortable and confident in navigating throughout the NJIT campus.

“I love how much this application has helped to ease navigation through campus!”

Bio

Cameron is in his first year at NJIT, studying Computer Science. He lives close to campus, which allows him to commute just in time for his classes. However, despite his proximity, Cameron often struggles to find his classrooms on time, leading to tardiness. Initially, his main challenge was navigating to the buildings themselves, but now he finds it equally challenging to locate specific classrooms within those buildings. This difficulty impacts his punctuality and adds unnecessary stress to his daily routine. Cameron is eager to find a solution that will facilitate his navigation around campus and help him arrive at his classes promptly.

Core needs

- Efficient and easy-to-use navigation to buildings and classrooms.
- Reminders for when classes are about to commence
- Personalized campus navigation based on schedule.

Frustrations

- Difficulty finding classrooms on time.
- Finds the on-campus map unhelpful, especially when stressed.
- Daily stress from navigating around campus each new semester.

Personal preferences

- Features catering to blind and colorblind individuals' needs.
- Integration with campus events calendar for reminders.
- Reminders for delays, cancellations, or classroom changes

Most Used Navigation App



Primary Device for Navigation



Mobile Phone

A low-angle, black and white photograph of several skyscrapers reaching towards the sky. The perspective creates a sense of height and scale. The buildings are dark, with some windows reflecting light. The sky is a uniform dark grey.

06

Problem Scenarios

Problem Scenarios

1. Brain is a second-year student majoring in computer science at NJIT. He has gotten used to finding classes but wishes the experience was easier. Certain buildings like the Architecture building are the most difficult to find classes in. They're comfortable using their email for campus events and warnings but could prefer a more streamlined notification system through an app. Brain isn't typically late for classes, but the time he has was because he forgot he had a class. In this case, a notification alert would have helped him, he admits. Brain believes he doesn't have a problem but wonders if an application can make his life easier.

2. Ray is a first year student. She is a chemistry major. She dorms but finds some buildings hard to understand. She went to a few classes late but because she got lost at certain places. She also felt like some of the elevators were really slow. For example the elevators in the library. She uses her often but not for classes because she feels like she doesn't need her phone for it. She also doesn't use her phone for events either. She also says that she doesn't use the maps on campus because it's really confusing.

Problem Scenarios Continued

3. Cameron, a first-year student at NJIT, struggles with navigating the campus, particularly finding his classes. Initially, what proved most challenging for him was deciphering building acronyms and locating them. However, now that some time in the spring semester has passed and he feels more comfortable navigating to these specific buildings, his main difficulty lies in navigating within them. Given his familiarity with Google Maps, he seeks a user-friendly application similar to it. However, he hasn't found such an application yet, leaving him anxious about future semesters and navigating new classes and buildings.

4. Carl is a second-year student who often finds it difficult to get to class on time because of the structure of the campus and he also finds it difficult that he can't find a well-organized place to find events so he either ends up missing them entirely or being late to them and he quotes "it makes me look bad." Carl is always used to being on time for his classes ever since high school so when he is greeted with the difficult layout of the NJIT campus he's overcome with anxiety and frustration and was often late to his first classes on his first year at NJIT and does not want the same thing to happen to other first-year students and he still finds some things difficult even though he is a second-year student.

A low-angle, black and white photograph of several skyscrapers reaching towards the sky. The perspective creates a sense of height and scale. The buildings are dark, with some windows reflecting light. A large, white, stylized number '07' is overlaid on the left side of the image. To the right of the '7', the words 'Activity' and 'Scenarios' are written in a white, serif font, stacked vertically. A thin white rectangular border frames the central text area.

07 Activity Scenarios

Activity Scenarios

1. Brian isn't usually late, but today there was Newark Traffic. He's going to be late for class, and his Technical Writing class announced that it's having a workshop at the Van Houtan Library. Brian takes out his trusty NJIT Campus Buddy and searches through the buildings. Once he arrives at Van Houtan Library, he selects the library in the app and looks for his room number. Using the map generated, he's able to swiftly locate the room without having to frantically search via the wall-mounted room numbers. Despite the traffic, he is able to make it to his workshop on time.

2. Ray dorms at NJIT but she has an event in the Faculty Memorial Hall on the fourth floor. She walks into Tiernan Hall and assumes both halls are connected. She gets his phone out and goes to the NJIT Campus Buddy and puts the locations she needs to go to and it shows a clear path redirecting her to walk into FMH (Faculty Memorial Hall). She made it on time to the event thanks to the NJIT Campus Buddy but there was a huge line because everyone heard free prizes. But when Ray got online she opened the NJIT Campus Buddy and had her QR code so when she got to the front she got in and got extra points to get Apple Music for free for the next month.

Activity Scenarios Continued

3. Come the next fall semester, Cameron has three classes, two of which are scheduled in new buildings to him, making him incredibly nervous since he hasn't had the opportunity to fully familiarize himself with the campus since the previous Spring semester. He remains confused by the abbreviations and locations of these buildings and classes as much as he was in his previous spring semester. However, upon taking a look at the on-campus map, he sees the mention of an application called “NJIT Campus Buddy,” provided to help students find their classes a lot faster, which causes a sense of relief to wash over him. He quickly downloads it, inputs his UCID and password, and is provided with navigation for his upcoming classes. With this straightforward procedure, he now feels confident that he will arrive at class on time and navigate the campus more efficiently in the future. He also appreciates that the app displays the abbreviations of each building alongside their full names, allowing him to memorize and associate each one.

4. Carl is a second-year student at NJIT and he often finds it difficult to navigate through campus during rush hour thankfully the NJIT Campus Buddy is an app that casually allows the student to navigate through the campus freely and Carl doesn't have to be late to class anymore because the app tells him an ETA of his time to class! However he is also in need of finding a quick and easy way to find events and to get there on time and NJIT Campus Buddy allows students to input their ID into the application and it will tell them all the things they need to know or do in order to get there on time because the app directly imports all of his classes and events into his calendar.

A low-angle, black and white photograph of several skyscrapers reaching towards a dark sky. The perspective creates a sense of height and scale. A white rectangular frame is superimposed over the center of the image.

08

Journey
Map

JOURNEY



Cameron

Scenario

It is Cameron's first day at NJIT, and he is on his way to the campus for the first time. He has his phone ready and open, with an image of his schedule ready to navigate confidently throughout the campus and to his upcoming classes.

Expectations

- Easy navigation throughout campus
- Easy to understand classroom layouts
- Arrival to classes on time

Phase 1

1. Arrives on campus.
2. Opens his phone to see where his first class is
3. Sees that it is located in "GITC", 1100

1
"Hopefully, this building and classroom will be easy to find."

Opportunity Point: Provide clear and concise directions to the first class location upon arrival.

Pain Point: Uncertainty about finding the building and classroom easily.

Phase 2

4. Was initially confused because he didn't see any buildings around him with that name
5. Realizes that he only has five minutes to get to class and begins to panic, running to the on-campus map to see where it is

2
"Where is the GITC building? I don't see it anywhere."

Opportunity Point: Offer quick and accurate navigation guidance to the GITC building, especially when time is limited.

Pain Point: Panic and confusion when unable to locate the GITC building on time.

Phase 3

6. Manages to discover the NJIT Campus Buddy through a QR code on the on-campus map.
7. Feels a bit more relieved knowing that this will guide him straight to class.

3
"All I need to do is input my ID to get directions to all my classes? I wish I knew about this earlier."

Opportunity Point: Discovering NJIT Campus Buddy through a QR code, providing relief and easy access to navigation assistance.

Pain Point: Regret over not knowing about the app earlier and experiencing the stress of navigating without it.

Phase 4

8. Successfully navigates to class just in time and, thanks to the app, is now confident he will be able to navigate to his upcoming classes.

4
"Thank goodness I discovered NJIT Campus Buddy"

Opportunity Point: Successfully navigating to class on time with confidence, thanks to the assistance provided by NJIT Campus Buddy.

Pain Point: Previous uncertainty and lack of confidence in finding classes efficiently.

Phase 5

9. Noticing his classmate arrive 10 minutes late to class, he decides to let him know about the NJIT Navigation app.

5
"This app is an easy 5 stars"

Opportunity Point: Sharing positive experiences with classmates and promoting the NJIT Navigation app, enhancing its visibility and reputation.

Pain Point: Observing classmates experiencing difficulties in navigation without the app's assistance.

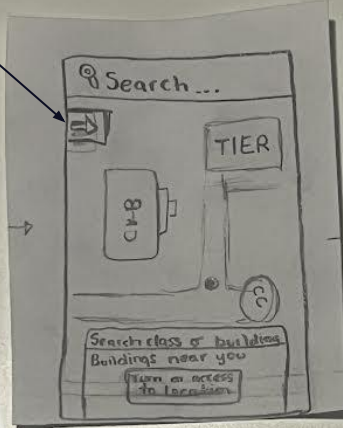
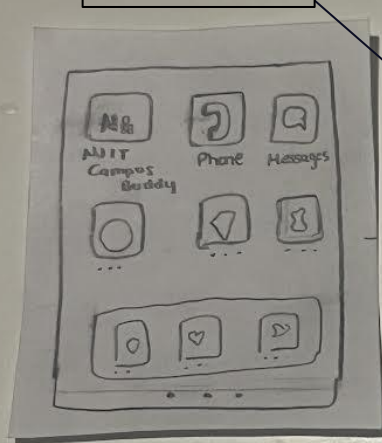
MAP

A low-angle, black and white photograph of several skyscrapers reaching towards a dark sky. The perspective creates a sense of height and scale. A white rectangular frame is superimposed over the center of the image.

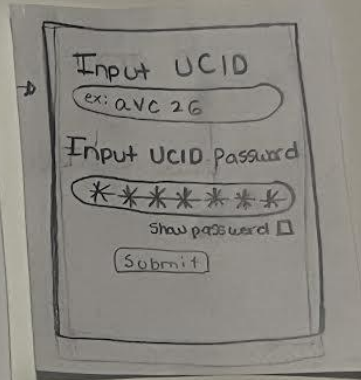
09 Paper Prototype

Task 1: Open the app and login

Lessons
Learned/New
Changes #1

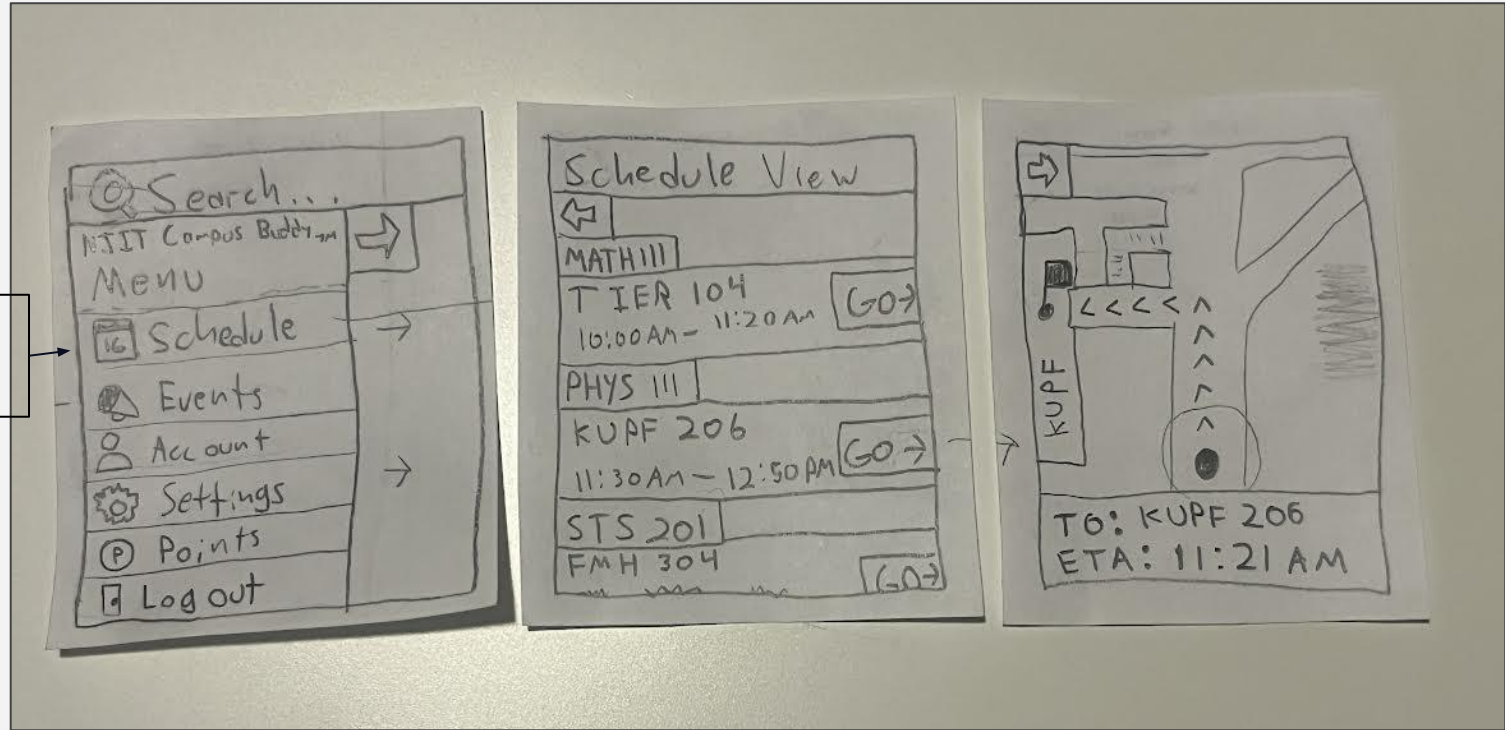


Lessons
Learned/New
Changes #3 [Screen
for logged-out user]

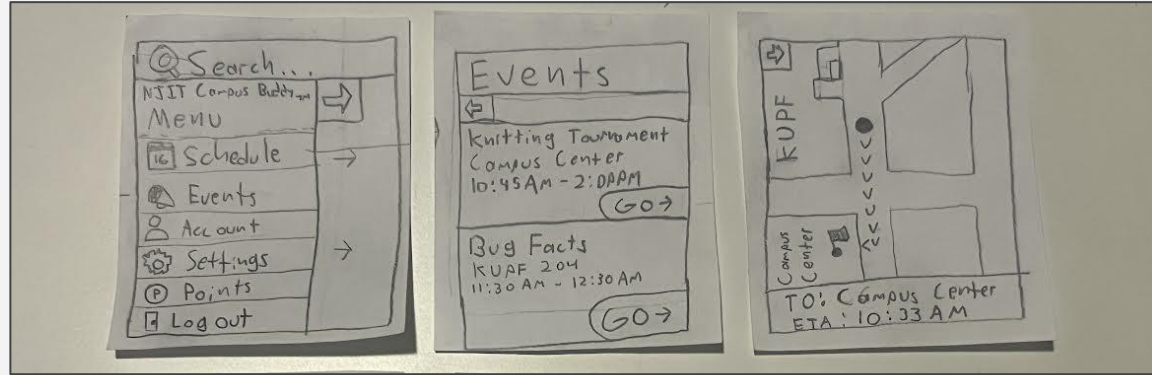


Task 2: Select an upcoming class to navigate to

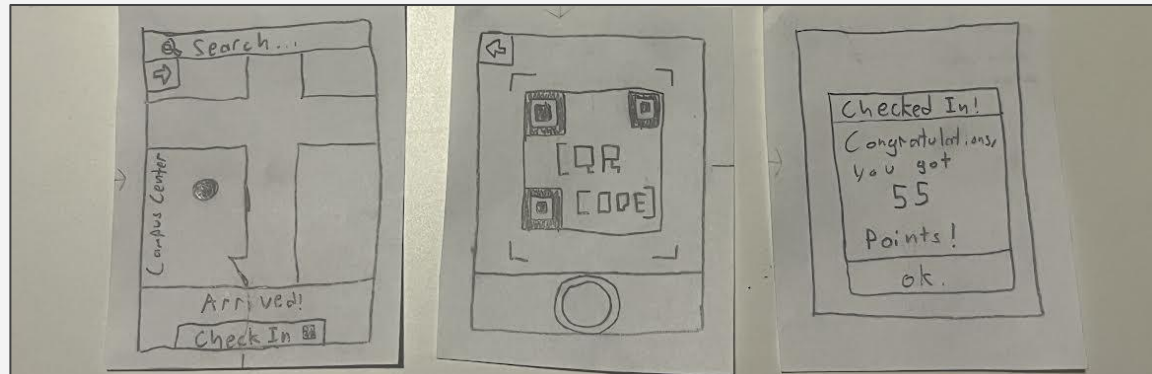
Lessons
Learned/New
Changes #2



Task 3: Select an upcoming event to navigate to

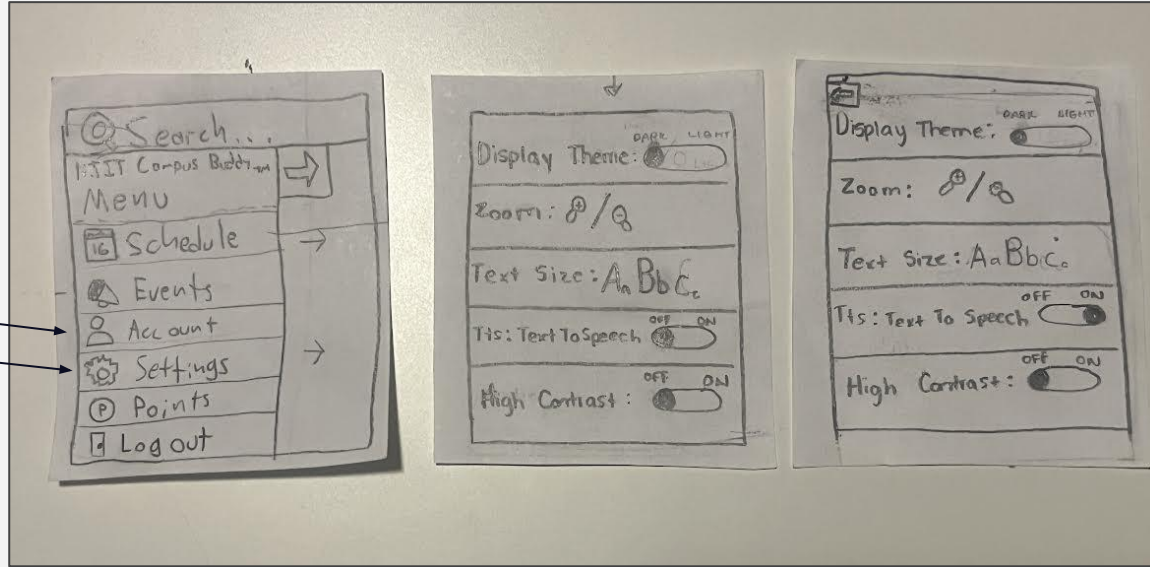


Task 4: "Use" QR code scanner to mark attendance and gain "points"



Task 5: Personalize your account's accessibility settings, such as enabling text-to-speech (TTS)

Lessons
Learned/New
Changes #3



A low-angle, black and white photograph of several skyscrapers reaching towards the sky. The perspective creates a sense of height and scale. The buildings are dark, with some windows reflecting light. The sky is a uniform dark grey.

10

Paper
Prototype
Evaluation
Results

Evaluation Results

User 1

Task 1: Not Done

Task 2: Not Done

Task 3: Done

Task 4: Done

Task 5: Not Done

User 2

Task 1: Not Done

Task 2: Not Done

Task 3: Done

Task 4: Done

Task 5: Done

User 3

Task 1: Not Done

Task 2: Not Done

Task 3: Done

Task 4: Done

Task 5: Done

Overall

Task 1: Inability to complete

Task 2: Inability to complete

Task 3: Easy to complete

Task 4: Easy to complete

Task 5: Both easy and unable to complete

Lessons Learned

1. A major issue we encountered when others were attempting to use our application was the difficulty in finding the “main menu”. Initially, we designed our menu as an arrow, hoping it would convey that pressing on it would open up a sidebar. However, this approach had the opposite effect and only served to confuse the users testing our paper prototype. [Task 1]
2. Another common issue was that our users seemed to be confused about where to click to find a class/event tailored to them for navigation. Although we initially believed that “Schedule” and “Events” was sufficient as a hint, we later realized that this was not the case. [Task 2]
3. Lastly, another common issue we encountered was locating the accessibility settings, which we initially placed separately in a “Settings” section. While 2 out of the 3 users intuitively found it, the user who did not highlighted how the “Account” section could also suggest an alternative area where personalization settings for accessibility can be found. This “Account” section was only meant to view points collected from attending events and logging in and out of NJIT Campus Buddy; however, it is understandable that the separation of the “Account” and “Settings” sections for finding accessibility settings, in particular, may be confusing. [Task 1, Task 5]

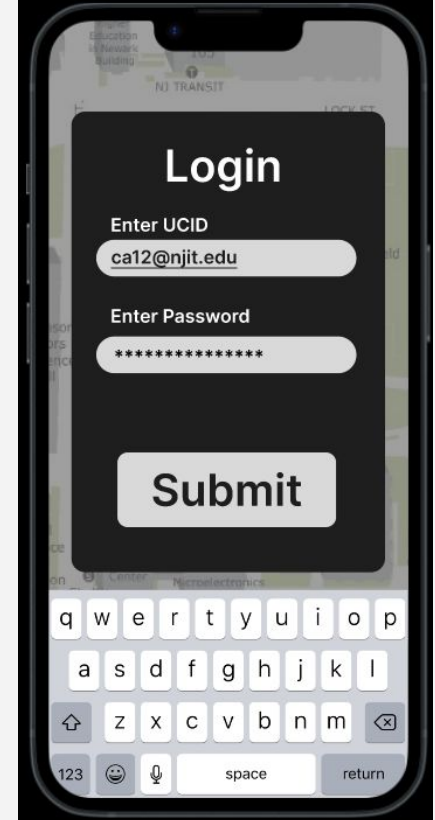
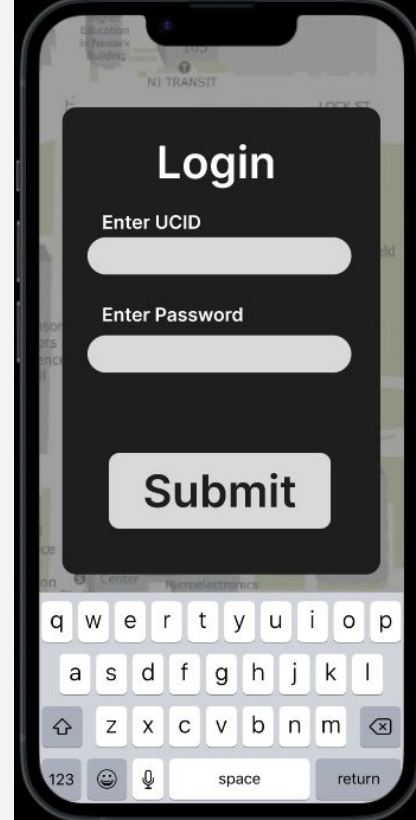
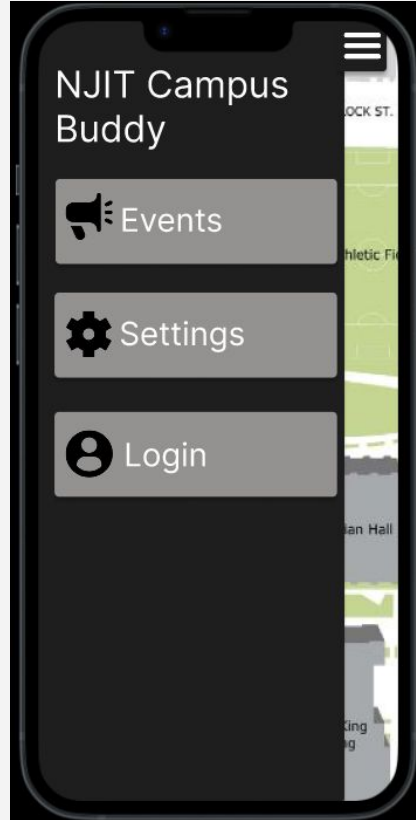
New Changes

1. Recognizing the difficulty users faced in finding the menu due to the arrow, we intend to enhance usability by adopting the hamburger sidebar method, a familiar navigation approach found in many other apps.
2. To address this, we'll rename the sections to “My upcoming classes” and “My upcoming events” to clarify that they contain the day's upcoming classes and events for users to access based on their personal account.
3. To address this issue, we will simply place the “Settings” section under the “Account” section, assuming that the user is logged in. However, the “Account” section will not appear when the user is not logged in, which would make it easier for the user to recognize the “Settings” section as a place to change accessibility features. During the paper prototype testing, the user happened to be logged in, which likely led to the confusion.

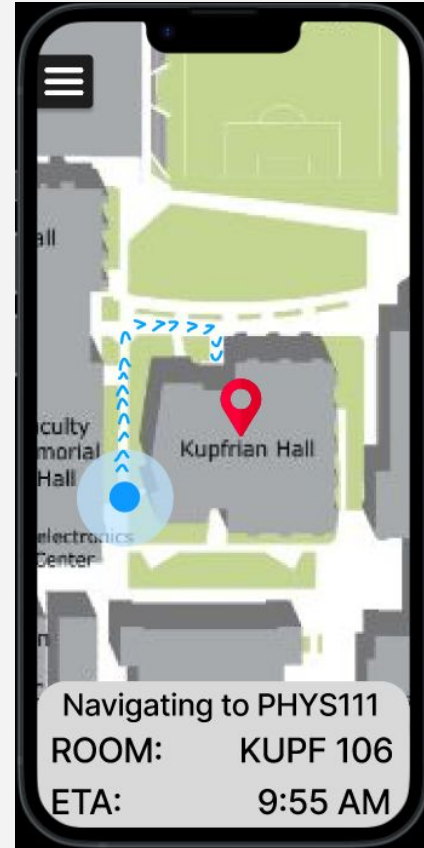
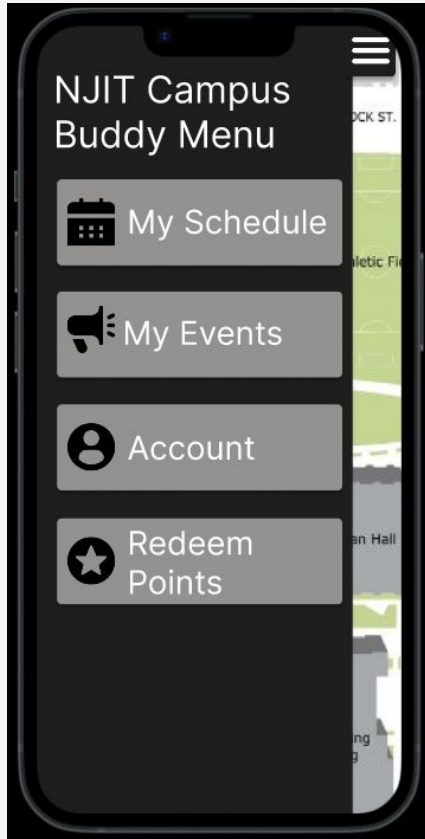
A low-angle, black and white photograph of several skyscrapers reaching towards a dark, cloudy sky. The perspective creates a sense of height and scale. A white rectangular frame is centered over the image, containing the text.

12 Medium Fidelity Prototype

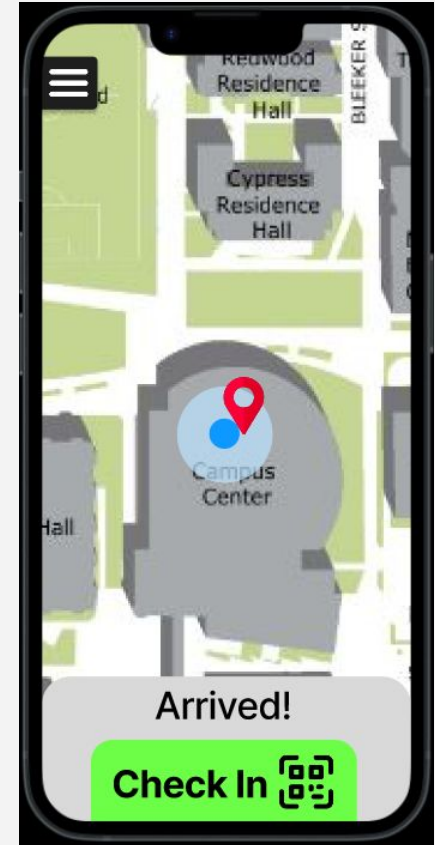
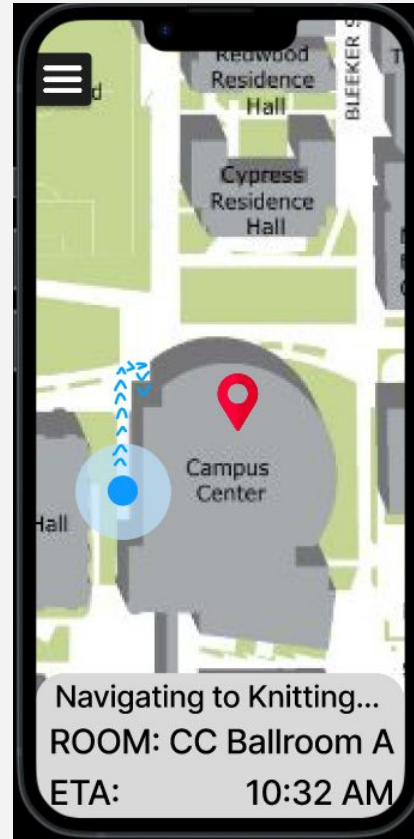
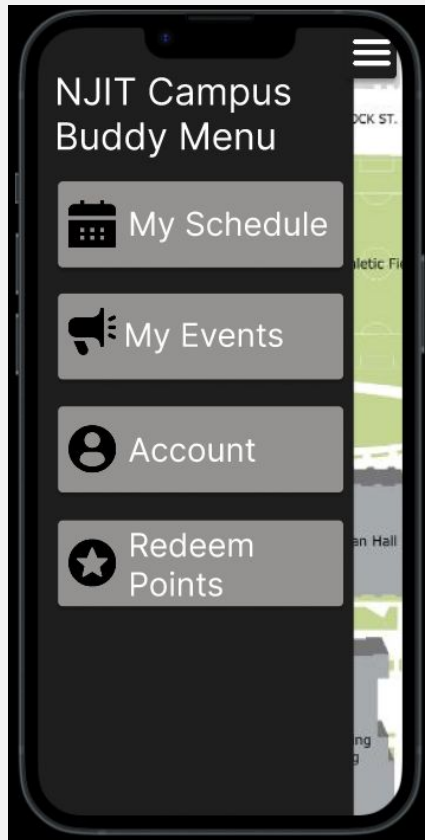
Task 1: Login to NJIT Campus Buddy



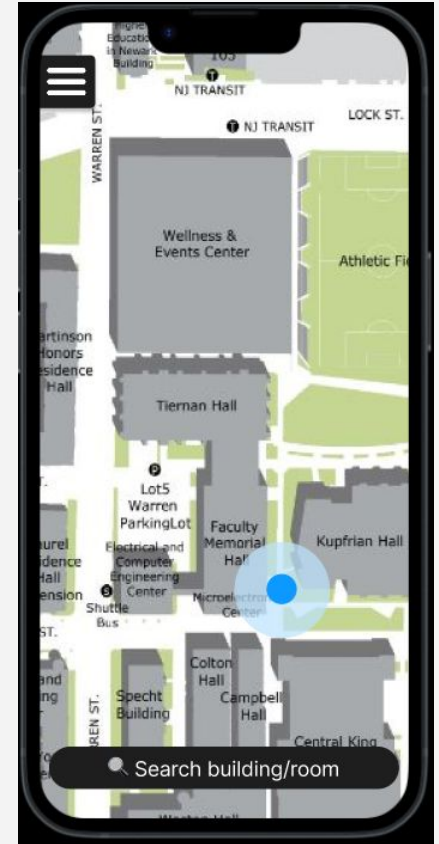
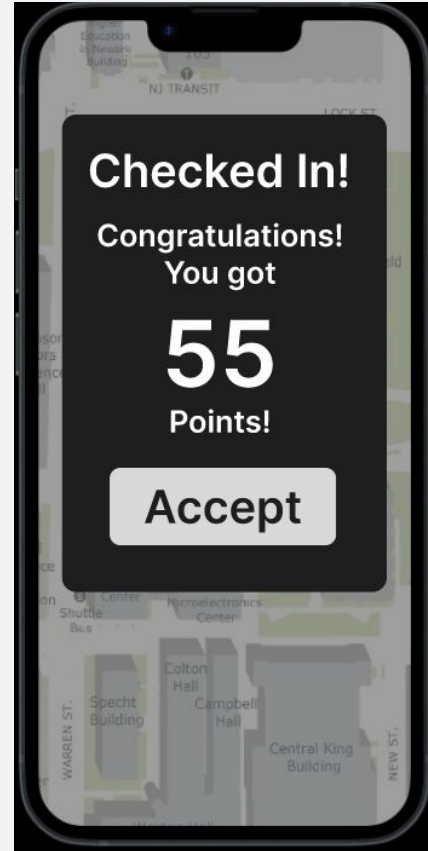
Task 2: Navigate to your Physics 111 class



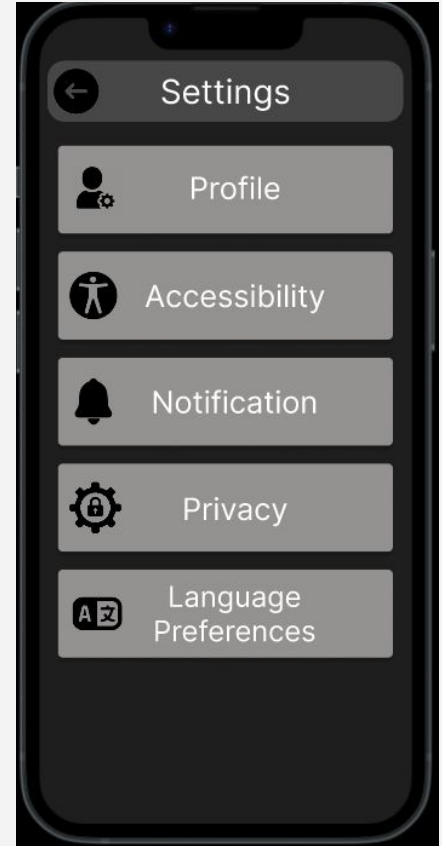
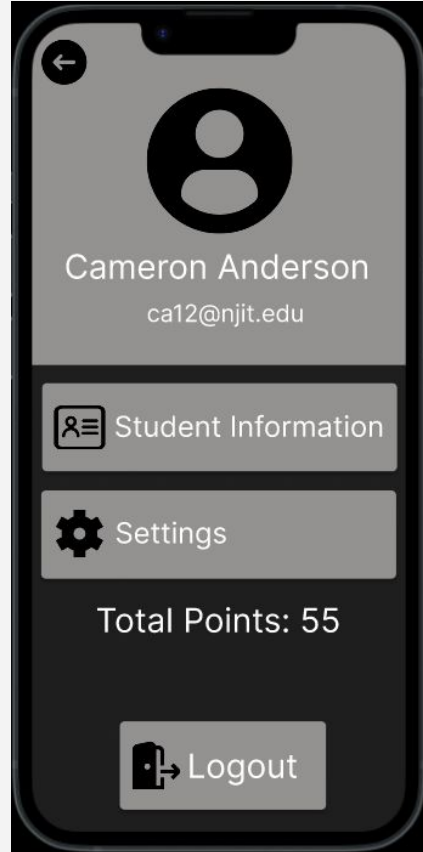
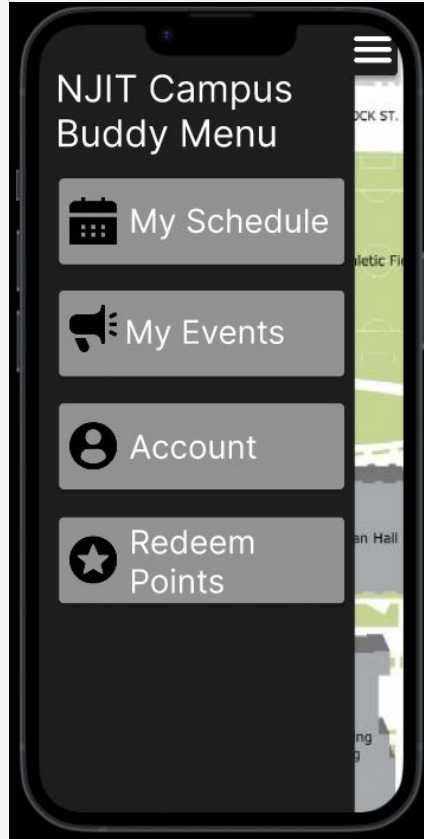
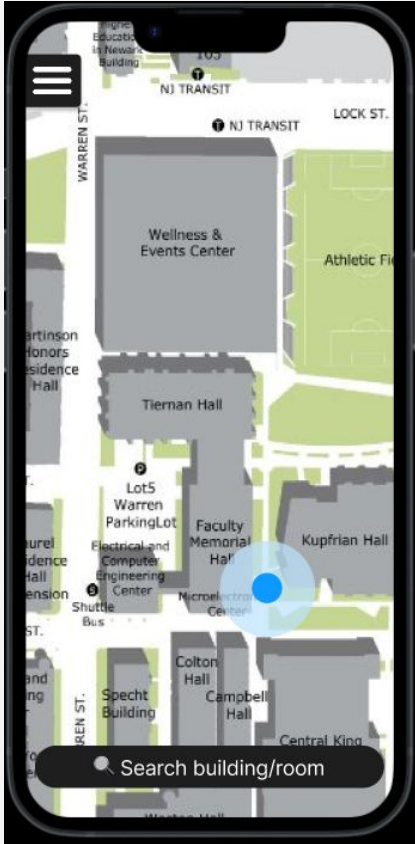
Task 3: Navigate to the Knitting Tournament



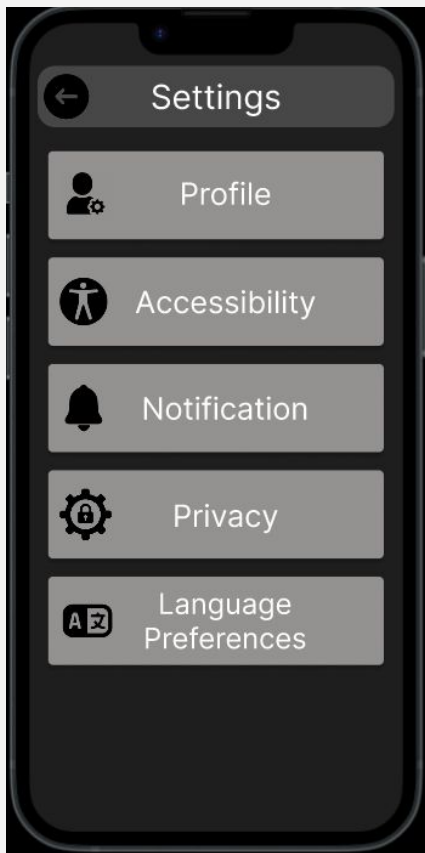
Task 4: Scan the QR code to receive points



Task 5: Personalize your account by turning on TTS: Text To Speech



Task 5: Personalize your account by turning on TTS: Text To Speech



Link to Figma
Prototype

**Additional
documents/sketches/artifacts**

A low-angle, black and white photograph of several skyscrapers reaching towards a dark sky. The perspective creates a sense of height and scale. A white rectangular border is centered on the page, framing the text.

01 Interview Questions

Interview Questionnaire

1. General Background:

- How long have you been at NJIT, and what's your major?
- Do you commute to campus, or do you live in the dorms?
 - If commuting:
 - Can you share a time when you were late to class while on campus, if ever?
 - If living in the dorms:
 - Can you recall a time you were late to class, if ever?

2. Navigation Experience:

- What are the challenges you encounter when trying to find your way to classes?
- Are there specific areas or buildings on campus that you find particularly confusing?

3. Current Navigation Solutions:

- Are there any existing apps or tools you find helpful for navigating the campus?
 - Can you tell me the last time you've used this application?
 - How was your experience?

4. Technology Usage:

- Are you comfortable using mobile apps for navigation?
- How often do you use your phone for campus-related activities?

Interview Questionnaire Continued

5. Preferences and Features:

- We are considering the development of an application that facilitates navigation within buildings, specifically for first-year NJIT students and potentially ongoing students, helping them locate classrooms more easily.
 - Do you think an application like this would be of value to you? Why/why not?
 - What features would you find most useful in a navigation app tailored for NJIT students?
 - What are things you would like us to consider when building something like this?
- How do you feel about receiving notifications or reminders about upcoming classes or events on your phone?
- Would you find it helpful to receive alerts about potential delays or changes in class locations?

6. Usability and Accessibility:

- How important is it for you that the app is easy to use and accessible?
- Are there any accessibility features that you believe should be included in the app?

A low-angle, black and white photograph of several skyscrapers reaching towards a cloudy sky. The perspective creates a sense of height and scale. A white rectangular frame is superimposed over the center of the image.

02

Affinity
Map

SAYS

Says that he loses track of the events that get emailed to me so he would find it convenient to receive notifications about upcoming classes.

Says he finds the Tiernan and Memorial hall confusing due to their layouts

Says the feature of receiving alerts about potential delays or changes in class location would be very useful

Says how he was late to class because he couldn't find his room in the CKB. He walked around the same floor 3 times only to realize that his class was missing it's room number.

User
1

Thinks that the NJIT Wifi is unreliable so it would be nice if you wouldn't have to be connected to the internet for the app to run well.

Thinks it is very important for the application to be easy to understand since his personal experience with google maps is poor due to lag/delay.

Thinks that our application would be useful not only for him in future semesters, but for several other new NJIT students.

THINKS

Uses navigation apps like google maps to find his way around bus stops.

rarely uses his phone unless it is for mandatory campus related activities

Feels that a text to speech feature would be important to add in order to account for any students with visual impairments.

Feels it would be nice to have a feature where you're able to send your location on campus to your friends (especially inside buildings) and vice versa, to find one another

DOES

Says he used the on campus map to find his way to buildings but had trouble finding his way to the actual classes in his first semester at NJIT

FEELS

SAYS

I have no trouble getting to class on time.

People who are late to class are lazy and disgust me.

I do not care about the Corq app

THINKS

The Architecture building is hard to traverse.

current navigation apps don't help properly navigate through campus

**User
2**

DOES

I dont use apps for anything on campus

uses email for notifications.

FEELS

Perfectly content with the status-quo

SAYS

I find Faculty Memorial Hall particularly confusing for the way that the rooms are numbered. It is not straightforward and I end up going up and down the halls to find

Yes I am comfortable with mobile apps for navigation. In fact my high school used to have a navigation app that helped get to classes because it was a big campus.

All i am aware of is the campus map that is placed outside but even those aren't that useful.

I am a first year freshman and this is the start of my spring semester. I major in chemistry.

THINKS

I believe we should consider the audience. There are blind individuals and something that could benefit them would be a voice saying what's on the screen out loud.

I would love to see updates on elevators, in order to see if they are broken or not working ahead of time.

I live on campus and I was just recently late to my calculus class last week. Sometimes i think i have more time then i actually do.

**User
3**

For campus activities I don't think I use my phone that often but in general I do spend a little of time on my phone.

Yes this would be very helpful and convenient especially if students don't have time to look at their emails or canvas.

I feel like this should be a key component because our campus is diverse and we have many different students in different situations. The app should be easy to use for everyone.

DOES

I would find this very useful especially a reminder for "travel time" so I won't be late.

I usually find that the elevators don't work or are slow and some of the numbering in the buildings are strange

FEELS

SAYS

The numbers of the rooms and the number of people during any time make it confusing to make it to class on time.

I find it useful to navigate myself through campus which would be public safety

I am comfortable using mobile apps for navigation

THINKS

There should be some accessibility features that could involve people's smartwatches because many people on campus use them, and it could be a good way to track or help

A feature that would be useful for a navigation app would be how to avoid traffic within-rush hour at NJIT.

**User
4**

commutes to campus.

doesn't use phone every day for campus-related activities.

feels as though it would be nice to be notified when there is an upcoming event or class on their phone

would find it very helpful to received alerts about potential delays or changes in class locations.

DOES

FEELS

SAYS

says it would be nice to see realtime status updates for things on campus.

says they would appreciate notification reminders on upcoming classes or events

says they're comfortable using mobile applications for navigation

THINKS

thinks certain buildings (Architecture, GITC, Tiernan, CKB) are difficult to navigate

thinks it would be helpful to receive notifications about potential delays or switches in classrooms

thinks a navigation app to find classes would be helpful

AGGREGATE
USER

Uses the Campus map outside most of the time

Knows the apps currently offered by NJIT, but don't use them very often.

Used to using navigation apps like google maps

feels as though it would be nice to see accessibility features for blind/colorblind people.

feels as though NJIT apps currently offered aren't enough (Corq)

feels that being shown the "travel time" to reach any class or building would be very helpful

DOES

FEELS

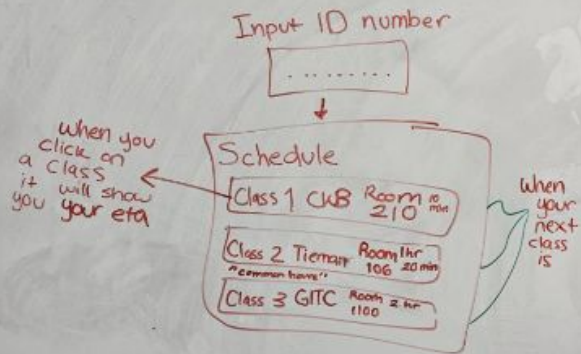
A low-angle, black and white photograph of several skyscrapers reaching towards a dark, cloudy sky. The perspective creates a sense of height and scale. A white rectangular frame is superimposed over the center of the image, containing the text.

03 Ideation Jamboard

Finalized User Goals

Goal One: To navigate to class with ease

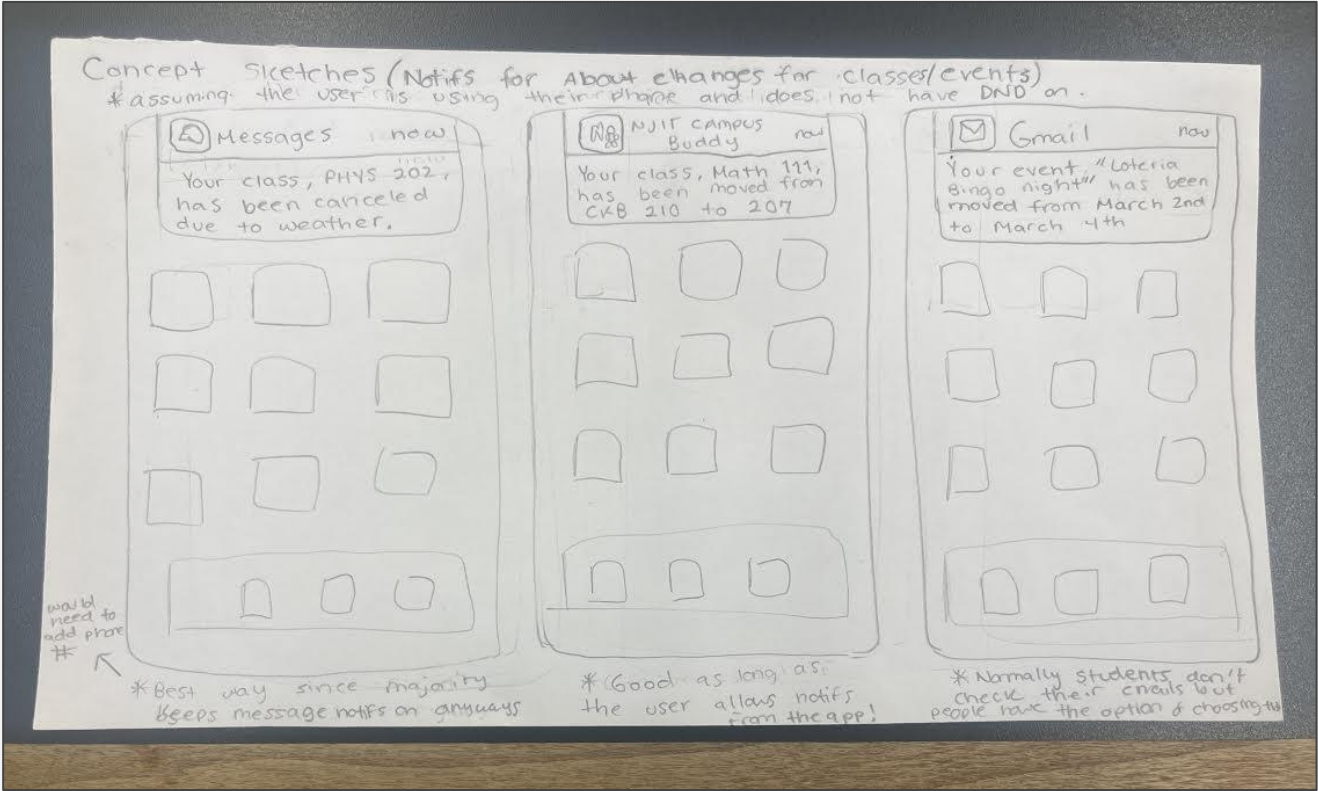
Goal Two: Be aware about abrupt changes from classes and events



A low-angle, black and white photograph of several skyscrapers reaching towards a dark sky. The perspective creates a sense of height and scale. A white rectangular frame is centered on the page, containing the title text.

04 Concept Sketches

Notification



Navigation

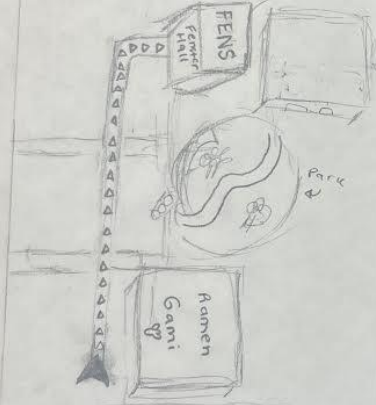
Concept sketches (Simple Campus Navigation)

IDEA 1



Continue straight

to Dr. Martin Luther King JR Blvd



To: Fenster Hall (FENS)

ARRIVING
3:05 PM



IDEA 2



Continue straight

to room 117

When at room
117, take a
left ← and

Current
time
3:02 PM

navigation
for
building



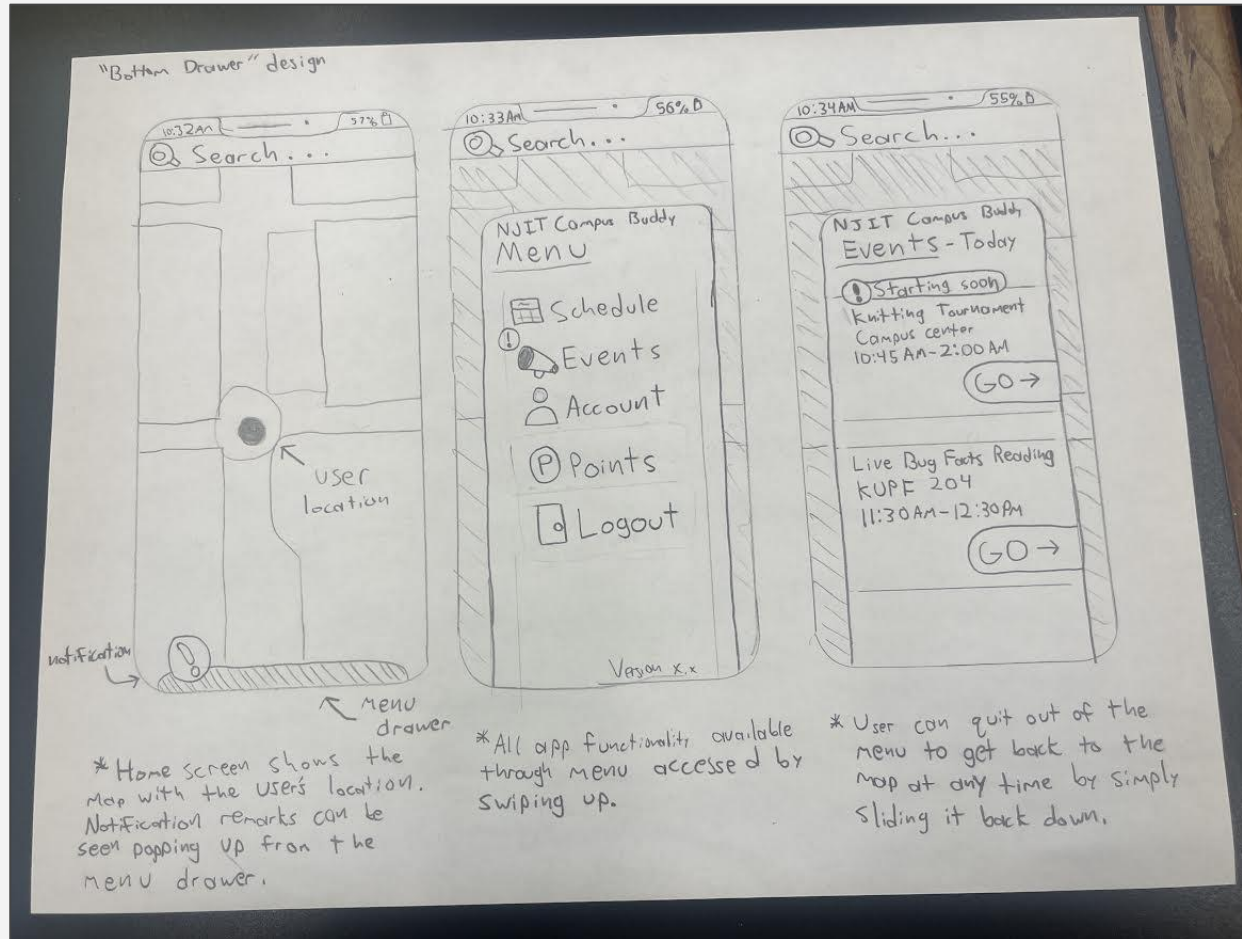
inside
look of
the
building

To: Central King Building (CKB)
Room 104

Arrival at
3:04 PM

Idea

Bottom Drawer



A low-angle, black and white photograph of several skyscrapers reaching towards a cloudy sky. The perspective creates a sense of height and scale. A white rectangular frame is superimposed over the center of the image.

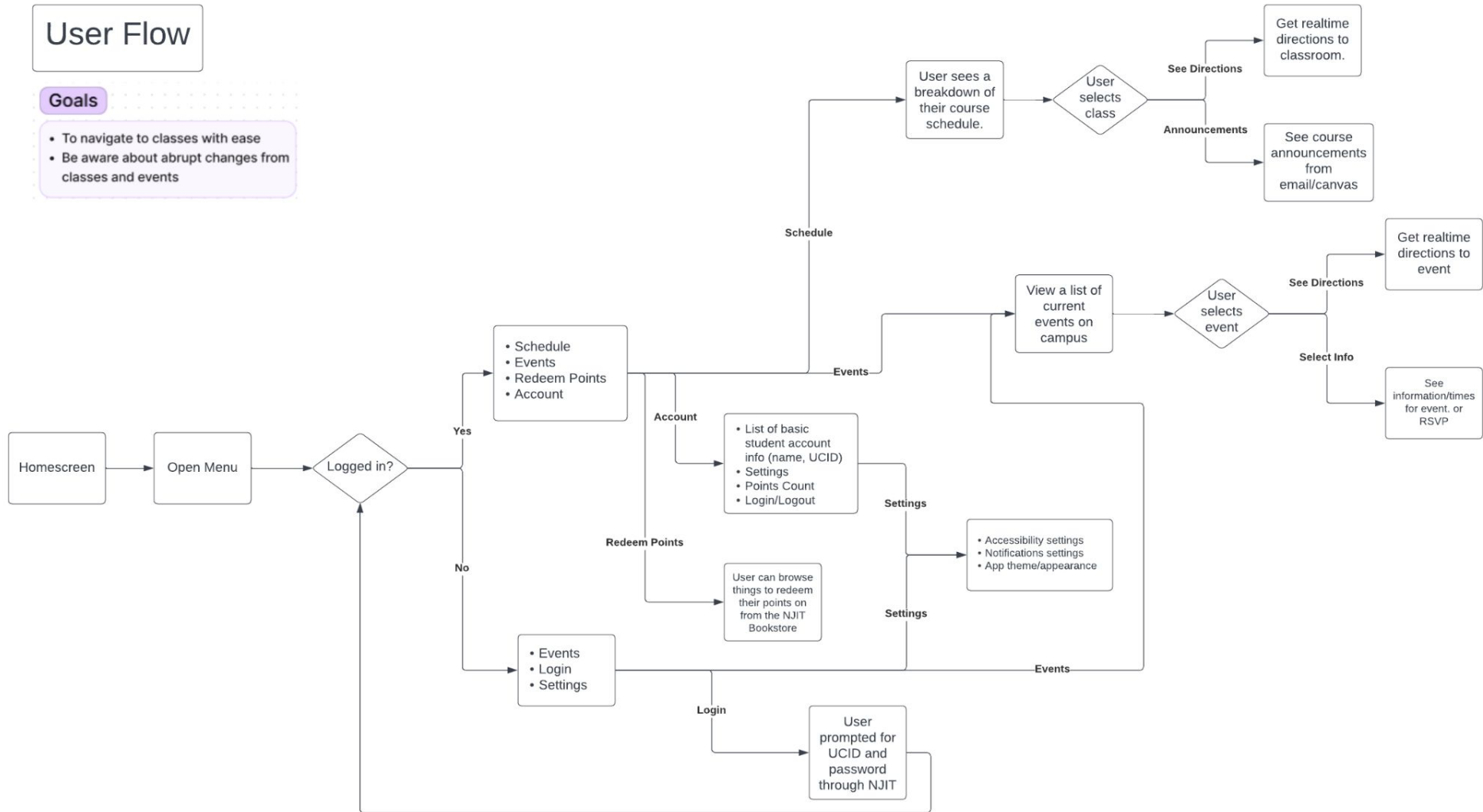
05

User Flow

User Flow

Goals

- To navigate to classes with ease
- Be aware about abrupt changes from classes and events



A low-angle, black and white photograph of several skyscrapers reaching towards the sky. The perspective creates a sense of height and scale. The buildings are dark, and the sky is a lighter, hazy grey.

06

Evaluation
Results of
Medium
Fidelity
Prototype

Results	Time Spent on Each Task [s]	Average Time Spent on Each Task [s]	Ease of Use [1-5] 1 = Difficult 5 = Easy
Task 1: Login to NJIT Campus Buddy	P1: 13	19	P1: 5
	P2: 25		P2: 3
	P3: 19		P3: 5
Task 2: Navigate to your Physics 111 class	P1: 18	17	P1: 5
	P2: 27		P2: 4
	P3: 6		P3: 5
Task 3: Navigate to the Knitting Tournament	P1: 19	About 12	P1: 5
	P2: 11		P2: 4
	P3: 8		P3: 5
Task 4: Scan the QR code to receive points	P1: 8	About 7	P1: 5
	P2: 9		P2: 5
	P3: 5		P3: 5
Task 5: Personalize your app by turning on TTS: Text-to-Speech.	P1: 24	About 18	P1: 5
	P2: 15		P2: 5
	P3: 17		P3: 5

During the evaluations, the majority of the participants quickly learned and adapted to the app. They knew exactly what they were doing during each test run after being given a task. However, in Tasks 1, 2, and 5, they took a bit longer than usual compared to Tasks 3 and 4. This was less about the app design and more about Figma's limitations, as overall ease of use in every task was nearly seamless. That said, it's important to note that there were still some errors present, unrelated to these limitations. Additionally, while not every suggestion a user makes should be implemented, one user suggestion stood out, which we believe as a group is worth integrating into our app.

Lessons Learned

1. In Task 1, much of the time was taken up by the screen where our login had a keyboard, as there was often an attempted use of the actual computer keyboard to log in by our users, unaware of Figma's prototype limitations.
2. In Task 2, the second user, who took more time than the others, clicked on 'my events,' but only because they didn't fully process that right above it there was a 'my schedule' option. Regardless, this error is important to take into consideration because it likely reflects an issue with our terminology, which is crucial in user design.
3. Task 3 went smoothly for all users.
4. Task 4 proceeded smoothly for all users. However, during Task 4, two users inquired about the purpose of the points after scanning the QR code. One of them suggested adding a help section to clarify this, which we considered a great idea, as many other users might have the same question. However, our approach to implementing this suggestion will be slightly different.
5. For Task 5, finding the settings within the account section was intuitive for each user, and a lot of time was spent looking through the options within the settings where text-to-speech would be found, which neither user made an error on, as they all clicked on the 'Accessibility' option. It is also worth noting that users who are disabled are more likely to intuitively click directly on this option, recognizing the common use of the icon.

New Changes

1. In Task 1, to account for this issue in our prototype and receive more accurate times, we will simply cut straight to the screen where the user hypothetically typed in their information.
2. For Task 2, while only one user ran into an issue due to moving too quickly, their error was worth noting because, intuitively, users should click on the correct option without hesitation even when navigating quickly. Thus, to address this, in the future, we will likely place both "My Schedule" and "My Events" in the same section under the main header of "My Upcoming Classes & Events" in our menu.
3. Task 3 went smoothly for all, and thus no changes will be made.
4. This change doesn't relate directly to Task 4 but rather to our entire application. However, it's the task where questions about our point system arose. To address this, in future development, we'll prioritize a quick tutorial for first-time users, explaining each section's purpose. This not only resolves the point system query but also addresses any other confusion. Additionally, we'll incorporate a ☐ Help section in our main menu for users who may need reminders.
5. For Task 5, due to the fact that most of the time was spent simply looking through the options to see which button would intuitively lead to an area where users can turn on Text To Speech and the fact that users who are disabled are more likely to click this option much faster since they're more used to it, no changes will be made to this portion either.

Conclusion

All in all, the results of this study helped to showcase that the changes implemented from the paper prototype made a drastic difference in the user flow. The only new changes that will be made this time around will be to ensure that our prototype takes into account Figma's limitations and to add a help section to further assist users in navigating through our app more efficiently, answering any likely questions that may come for all menu options.